

Entity Resolution, Identity Collapse, and the Drifted Self

How Systems Turn Fragmented Records Into an Operational Person

Knowledge Systems and Reality Drift - Note #6 v1

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The Basic Pattern

Large information systems rarely encounter a person or organization directly. They encounter records: names, addresses, identifiers, transactions, accounts, documents, and mentions. Entity resolution determines which of these records refer to the same real-world entity.

The task is essential. Without it, one person may appear as several customers, patients, authors, or risks. But resolution also creates a canonical identity by deciding which fragments belong together and which must remain separate.

A false merge combines different entities. A false split divides one entity into several representations. Between them lies a more subtle problem: even a technically correct match can produce an operational identity that emphasizes what is recorded and suppresses what is not.

When the Canonical Entity Becomes the Person

Entity resolution uses identifiers, attributes, contextual relations, similarity scores, and probabilistic rules. The resulting entity is not the person. It is a maintained representation assembled for a particular purpose.

Once records are linked, information travels across them. A risk flag from one account may affect another. A mistaken address may become evidence of association. A historical classification may follow the resolved entity into new contexts.

The canonical record gains authority because it appears comprehensive. People are asked to prove that the representation is wrong, even though they did not control the records, matching rules, or confidence thresholds through which it was assembled.

How Identity Collapse Emerges

Stage 1 - Fragmentation

A real entity appears across multiple records, sources, spellings, identifiers, and contexts.

Stage 2 - Resolution

Rules or models calculate which records likely refer to the same entity.

Stage 3 - Canonicalization

Matched records are consolidated into a canonical node, profile, or identity.

Stage 4 - Propagation

Attributes, relations, and judgments travel across the merged representation.

Stage 5 - Identity Collapse

Roles, time periods, sources, and probabilistic associations lose their boundaries inside the canonical profile, allowing information from one context to govern another.

Identity Collapse and the Drifted Self

Entity resolution creates continuity by binding fragmented records into a persistent identity. Identity Collapse begins when that continuity erases distinctions the system still needs to preserve: distinctions between roles, contexts, time periods, sources, verified facts, and probabilistic associations.

A person may correctly be identified across several databases while still being misrepresented by the identity assembled from them. A temporary address, historical classification, shared device, disputed transaction, or inferred relationship can migrate across the canonical entity and acquire consequences far beyond its original context.

The Drifted Self is the broader condition that can result. It is the operational version of a person that persists across databases, platforms, models, and institutional records while becoming progressively less answerable to the changing person it represents. Entity resolution does not create this condition by itself, but it can accelerate it by joining previously separate traces and allowing judgments to propagate across them.

The system gains continuity. The person loses contextual boundaries. Actions taken in one role become evidence about another. Past conditions become present attributes. Inferred connections begin functioning as established features of identity.

A reality-constrained identity system must therefore preserve uncertainty, provenance, temporal boundaries, contextual separation, and appeal. The canonical entity should remain decomposable. Users and decision-makers need to see which records produced it, which links were inferred, and which later judgments depend on those links.

The Representation Stack

A person produces traces. Traces become records. Records are matched into an entity. The entity becomes a profile. The profile becomes a score, category, or decision surface. The further the representation travels, the less visible the original matching decision becomes.

A single false link low in the stack can therefore reorganize many later conclusions. Correction requires more than changing one field. It requires tracing and revising every inference that depended on the resolved identity.

Examples Across Systems

Healthcare: Records from different patients can be merged, while one patient's fragmented records can conceal a complete clinical history.

Credit and Fraud Systems: Shared addresses, devices, or names can create inferred associations that become operational risk signals.

Scholarly Knowledge Graphs: Authors with similar names may be merged, while one author publishing under variants is split across profiles.

Consumer Platforms: Accounts, devices, purchases, and browsing traces are linked into an identity used for personalization and prediction.

Conclusion

Entity resolution does not simply discover identity. It constructs an operational answer to the question of which records count as the same thing. That answer can be useful, probabilistic, revisable, and still profoundly consequential.

The safeguard is keeping the canonical entity answerable to the person, organization, and evidence from which it was assembled. A resolved identity should never become an indivisible fact.

Related Phrases and Concepts

- entity resolution
- record linkage
- entity disambiguation
- identity resolution
- deduplication
- canonical entity
- false merge
- false split

Semantic Fidelity Project Resources

- [Research Library \(GitHub\)](#)
- [Articles & Essays \(Substack\)](#)
- [Semantic Fidelity Glossary](#)